Threat Model: Bellevue School District ParentVUE System

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1. Executive Summary

The Bellevue School District (BSD) ParentVUE system provides parents, teachers, and administrators with secure access to student information, including grades, attendance, health records, and communication tools. Because the system handles sensitive student data protected under FERPA, a structured security assessment was conducted using Data Flow Diagrams (DFD Level 0 and Level 1), the STRIDE threat-modeling framework, and a qualitative risk matrix.

The assessment identified several high-risk threats, including credential theft, unauthorized access to student records, and potential database compromise. Medium-high risks include tampering with grades, messaging impersonation, and denial-of-service attacks targeting authentication services. A comprehensive mitigation strategy was developed, incorporating multi-factor authentication, encryption, RBAC, secure coding practices, and availability protections.

This report provides a complete threat model, risk analysis, and mitigation plan to strengthen the confidentiality, integrity, and availability of the ParentVUE system.

2. Introduction

ParentVUE is a critical component of BSD's digital ecosystem, enabling families to stay informed and engaged in their children's education. Because the platform handles sensitive personal and academic information, a formal threat model is necessary to ensure the system remains secure, compliant, and resilient. This report evaluates ParentVUE using industry-standard methodologies and provides actionable recommendations to enhance system security.

3. System Overview

BSD ParentVUE is a web and mobile platform that allows parents to access report cards, review daily attendance, message teachers, and update student details such as demographics, emergency contacts, addresses, and health information. The system integrates with the Synergy Student Information System (SIS), which stores all district student data.

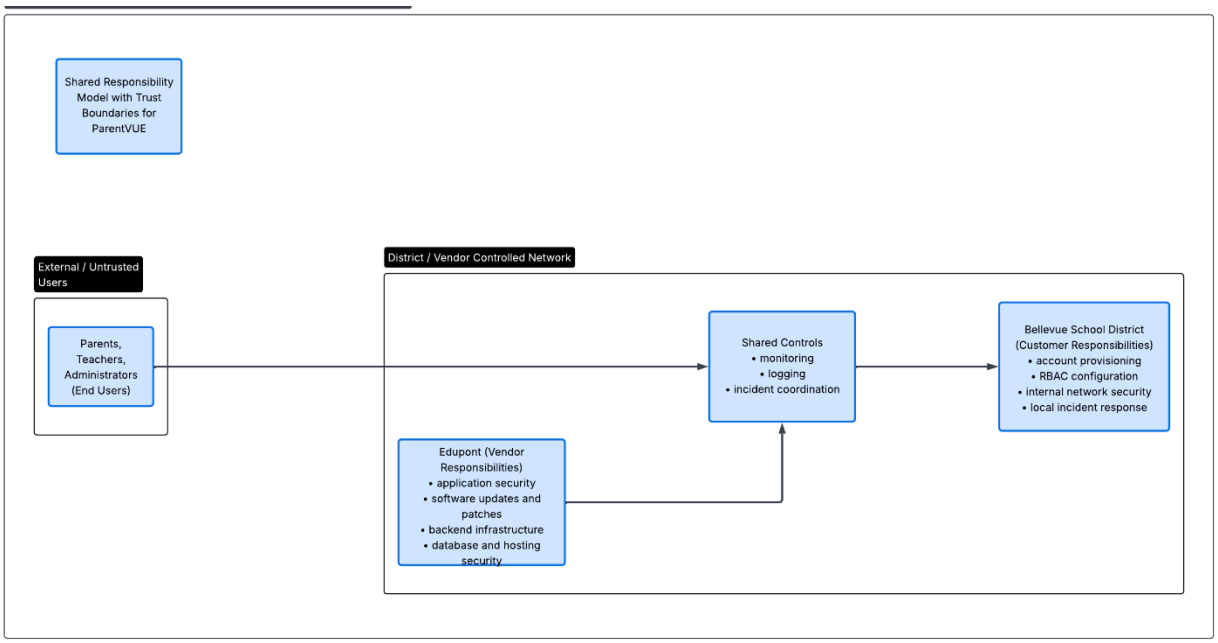
4. Key Stakeholders

The ParentVUE system involves several stakeholders, including parents, teachers, school administrators, district IT staff, and Edupoint as the software vendor. Each group plays a specific role in accessing, managing, or supporting the platform.

Vendor and District Security Responsibilities: ParentVUE follows a shared responsibility model between Edupoint and the Bellevue School District. Edupoint secures the ParentVUE/Synergy platform, including software updates, patches, and backend infrastructure. The district manages user accounts, RBAC settings, internal network security, and local incident response. Both parties share monitoring and coordination duties, which is essential for accurate threat modeling and risk assessment.

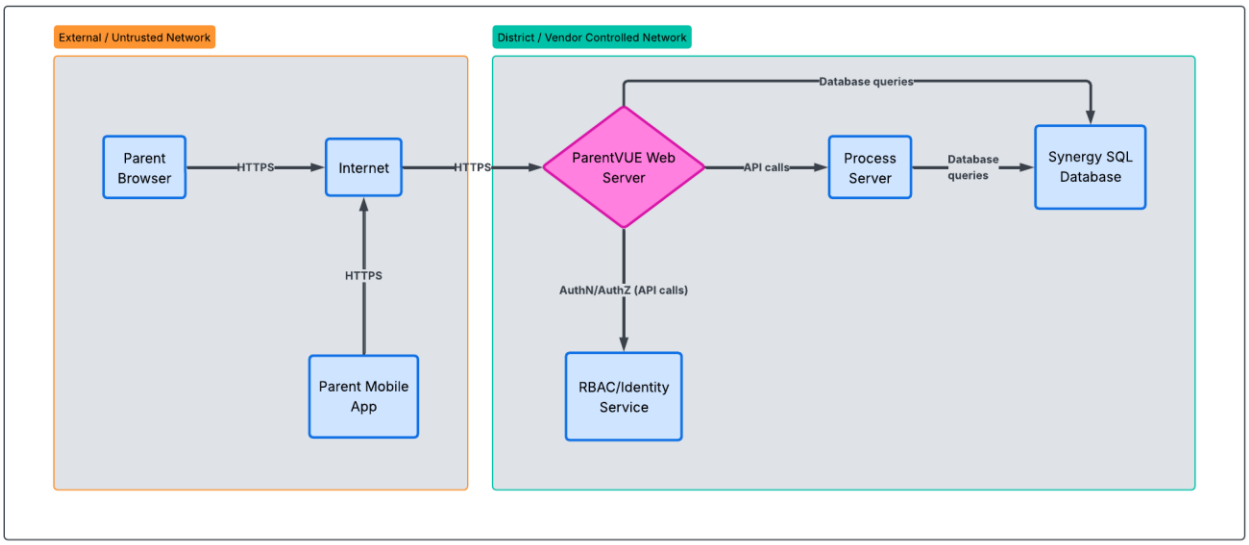
Parents: Primary users who access student grades, attendance, and update information.
Teachers: Enter grades, attendance, and communicate with families. School Administrators: Manage enrollment, student records, approvals, and data accuracy.
BSD IT and Security Teams: Maintain identity management, system configuration, and security controls.
BSD Leadership: Set policies, compliance requirements, and oversee system governance.
Edupoint (Vendor): Hosts the ParentVUE/Synergy platform, maintains backend services, and provides software updates.
State Education Agencies: Rely on accurate reporting and compliance with FERPA and state data rules.

4.1. Figure 1: Shared Responsibility Model with Trust Boundaries for the ParentVUE System.



5. High-Level Architecture Overview

5.1. Figure 2: Architecture Diagram



Presentation Layer (Interfaces)

Web Portal: Parents access ParentVUE through the district's portal. Mobile Apps (iOS/Android): Native apps communicate with the same backend APIs as the web portal.

Application Layer (Logic)

Web Server: Manages traffic from browsers and mobile devices. Process Server: Handles high-resource tasks such as generating PDF report cards or transcripts.

Data Layer (Storage)

SQL Database Server: Stores all district student data. Role-Based Access Control (RBAC): Ensures that user roles strictly define what data can be read or edited.

6. Assets

- ParentVUE web portal
- ParentVUE mobile app
- Parent credentials (usernames, passwords, activation keys)
- Student personal information
- Academic records

- Attendance logs and schedules
- Health information
- Synergy Mail messaging
- District reputation
- District infrastructure

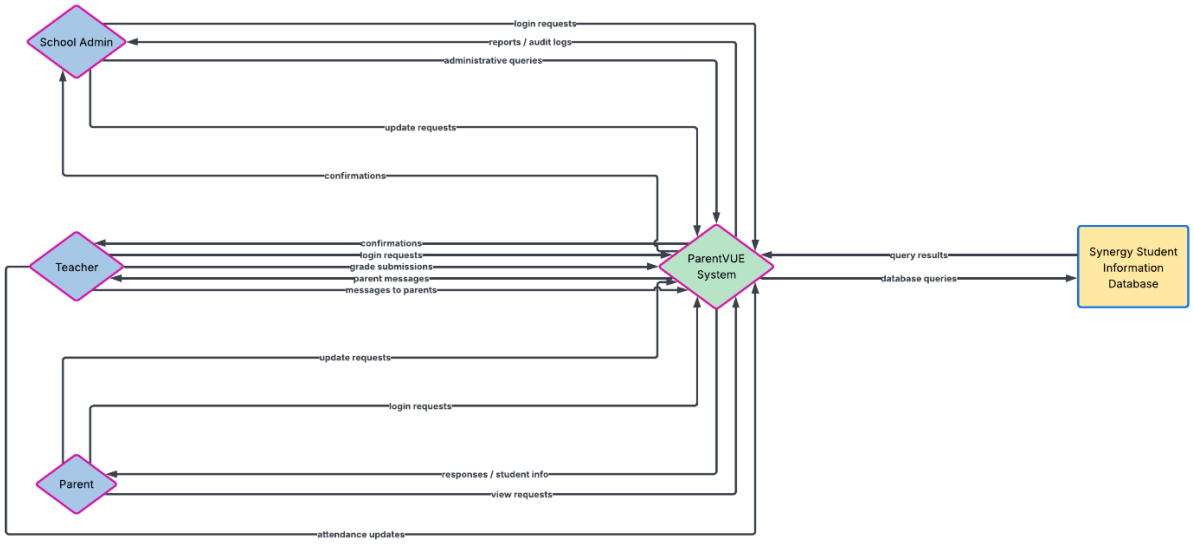
7. Trust Boundaries and Data Flows

ParentVUE includes several trust boundaries:

External Boundary: Parents, teachers, and administrators accessing the system from personal devices. Internal District Boundary: Web servers, process servers, and the Synergy database. Vendor Boundary: Edupoint's hosted backend services.

8. Data Flow Diagrams

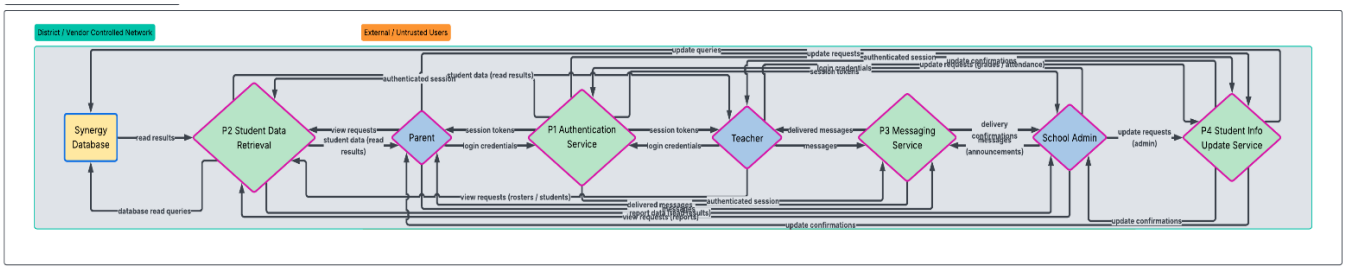
8.1. Figure 3: DFD Level 0



The DFD Level 0 diagram provides a high-level view of how external users—parents, teachers, and school administrators—interact with the ParentVUE system. It shows the primary data flows entering and leaving the system, including login requests, grade and attendance inquiries, messaging, and student-information updates. This context diagram establishes the system

boundary and illustrates how ParentVUE communicates with the Synergy Student Information System (SIS) as the central data store.

8.2. Figure 4: DFD Level 1



The DFD Level 1 diagram expands the ParentVUE system into its internal processes, including authentication, student-data retrieval, messaging, and information display. It illustrates how user requests move through these processes and how each component interacts with the Synergy Student Information System (SIS) to retrieve or update data. This diagram provides a more detailed view of system behavior and highlights the trust boundaries where security controls must be enforced.

9. Threat Modeling Framework Selection

STRIDE was selected because it aligns with the types of threats relevant to ParentVUE, including authentication, data retrieval, messaging, and interactions with the Synergy SIS backend. STRIDE provides a structured method for identifying threats across Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege.

10. STRIDE Threat Model

The STRIDE analysis identified threats across external entities, internal processes, data flows, and the Synergy database.

Table 1. STRIDE Threat Analysis for the ParentVUE System

STRIDE Category	Threat Description	Affected Components	Impact	Recommended Mitigations
Spoofing	Stolen or guessed parent credentials allow attackers to	Login Page, Authentication	Unauthorized access to student	MFA, strong password policy, account lockout,

STRIDE Category	Threat Description	Affected Components	Impact	Recommended Mitigations
	impersonate legitimate users.	Service (P4), Parent Accounts	records; privacy violations	secure session tokens, login anomaly monitoring
Spoofing	Compromised teacher/admin accounts used to impersonate staff.	Messaging Service (P3), Admin Accounts	False communication, misinformation, reputational damage	MFA for staff, RBAC enforcement, device-based authentication
Tampering	Unauthorized modification of grades, attendance, or student information.	Student Data Retrieval (P1), Synergy Database	Data integrity loss; academic manipulation	Server-side validation, audit logs, RBAC, database integrity constraints
Tampering	Alteration of messages sent through Synergy Mail.	Messaging Service (P3)	False messages, misinformation, trust issues	Message signing, TLS, input validation
Repudiation	Users deny actions such as updating student info or sending messages.	Web Portal, Mobile App, Synergy Mail	Disputes, lack of accountability	Audit logging, timestamping, non-repudiation controls
Information Disclosure	Exposure of PII during transmission (e.g., intercepted traffic).	Client ↔ Web Server ↔ Database Data Flows	FERPA violations, privacy breaches	TLS 1.2+, encryption at rest, secure API endpoints
Information Disclosure	Over-privileged accounts accessing unauthorized data.	RBAC System, Synergy Database	Unauthorized data exposure	Least privilege, RBAC review, access-control audits

STRIDE Category	Threat Description	Affected Components	Impact	Recommended Mitigations
Denial of Service	Flooding authentication endpoints to block logins.	Authentication Service (P4), Web Server	System unavailability	Rate limiting, traffic throttling, load balancing
Denial of Service	Excessive requests for large PDF reports overload the Process Server.	Process Server	Performance degradation	Queue management, resource throttling, caching
Elevation of Privilege	Attackers escalate from parent to teacher/admin roles.	RBAC, Web Server, Database	Full system compromise; unauthorized data modification	Strict RBAC enforcement, privilege separation, server-side authorization
Elevation of Privilege	Exploiting API vulnerabilities to bypass role checks.	Backend APIs, Web Server	Unauthorized access to restricted data	API gateway validation, input sanitization, penetration testing

High-Risk STRIDE Findings

Spoofing: Credential theft, account impersonation Tampering: Unauthorized modification of grades or attendance Information Disclosure: Exposure of PII during transit or storage Denial of Service: Flooding authentication or messaging endpoints Elevation of Privilege: Unauthorized escalation to administrative roles.

Impact on STRIDE and Risk Assessment: The shared responsibility model determines which threats belong to Edupoint and which belong to the Bellevue School District. Platform and backend vulnerabilities fall under Edupoint, while account misuse, RBAC issues, and internal network risks fall under BSD. Shared threats—like credential theft or suspicious logins—require coordinated monitoring and response. This division is reflected in the STRIDE table and risk ratings.

11. Risk Assessment

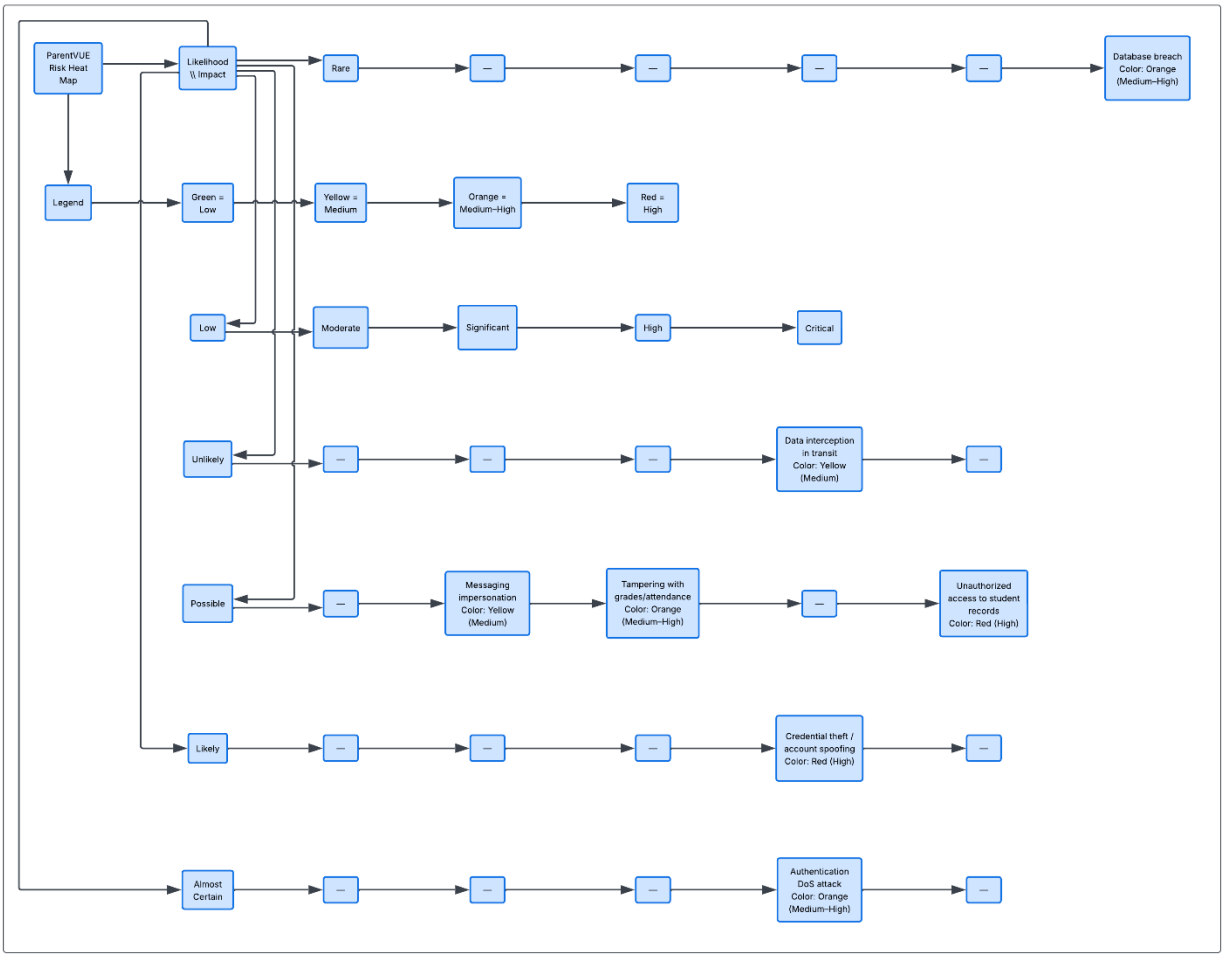
11.1. Risk Matrix / Heat Map

The risk matrix illustrates how threats are distributed across likelihood and impact categories. High-risk items appear where “Likely” or “Possible” threats intersect with high or critical impact, indicating areas requiring immediate mitigation. The heat map visually highlights these concentrations, helping prioritize security controls.

Table 2: Risk Assessment Matrix

Likelihood \ Impact	Low	Medium	High	Critical
Rare	Low	Low	Medium	Medium
Unlikely	Low	Medium	Medium	Medium-High
Possible	Medium	Medium-High	High	High
Likely	Medium-High	High	High	Critical
Almost Certain	High	High	Critical	Critical

Heat Map



11.2. Key Risks Identified

The key risks table summarizes the most significant threats to the ParentVUE system, including credential theft, unauthorized access to student records, and database compromise. These risks are rated based on severity and potential impact on confidentiality, integrity, and availability. Medium-high risks such as messaging impersonation and authentication DoS attacks also require proactive mitigation to maintain system reliability and trust.

Table 3: Key Risks Identified

Risk	Description	Severity	Reasoning / Impact
Credential Theft / Account Spoofing	Attackers steal parent or staff credentials to access student data.	High	Leads to unauthorized access to PII; FERPA violation; high likelihood.

Risk	Description	Severity	Reasoning / Impact
Unauthorized Access to Student Records	Exploiting weak RBAC or compromised accounts.	High	Exposure of sensitive academic, demographic, and health data.
Database Breach	Compromise of Synergy SIS backend.	High	Catastrophic data loss, manipulation, or mass disclosure.
Grade or Attendance Tampering	Unauthorized modification of student records.	Medium-High	Impacts data integrity and academic trust.
Messaging Impersonation	Sending fraudulent messages via Synergy Mail.	Medium-High	Misinformation, reputational harm, parent–teacher trust issues.
Authentication DoS Attack	Flooding login endpoints to block access.	Medium-High	Prevents parents and staff from accessing critical information.
Data Interception in Transit	Sniffing or intercepting unencrypted traffic.	Medium	Privacy risk, but mitigated by TLS when properly configured.

The assessment identified several high-priority risks that pose significant threats to the confidentiality, integrity, and availability of the ParentVUE system. **High-risk threats** include credential theft or account spoofing, unauthorized access to student records, and potential database compromise—each of which could result in severe privacy violations and FERPA non-compliance. **Medium-high risks** include tampering with grades or attendance, messaging impersonation, and denial-of-service attacks targeting authentication services, all of which could disrupt system reliability or undermine trust between families and the district. **Medium-level risks**, such as data interception in transit, remain important but are effectively mitigated through encryption and secure communication protocols.

12. Mitigation Summary

Authentication Security: MFA, strong passwords, secure session tokens Data Protection: TLS encryption, encryption at rest, data minimization Access Control: RBAC, least privilege, server-side authorization Integrity Controls: Input validation, message signing, audit logging Availability Protections: Rate limiting, load balancing, anomaly detection Error Handling: Sanitized error messages

These controls support a defense-in-depth strategy.

13. Conclusion

The ParentVUE system plays a vital role in supporting communication and transparency between BSD and families. This assessment provides a comprehensive understanding of system risks and actionable recommendations to enhance security. By aligning controls with actual data flows and trust boundaries, BSD can better protect sensitive student information and maintain public trust.

14. Author Contribution Statement

Copilot supported the writing process, while I completed the system analysis, diagrams, and security evaluation myself.

15. References

1. https://copilot.microsoft.com/
2. https://www.lucidchart.com/